DELPH-IN Applications in Second Language Education:

A semantic-based MT approach to ambiguity resolution

- **PhD Proposal & Proof of Concept MOE TRF**
- **Inspired by work done for the ERG** + few twists:
  - **Multilingual** (English, Chinese & Japanese)
  - **Targeting University Level L2 Learners**
  - **Integrated with WNs** (gradual introduction of vocabulary)
  - …



No Big News: Language is Ambiguous

Student:	That dog like the cat happy.
	Hmm… something is wrong with your sentence. Did you mean any of these? **A.** 那只狗和猫一样高兴。 [That dog, like the cat, is happy.]* **B.** 那只狗喜欢猫高兴。 [That dog likes the cat happy.]* **C.** 那只狗喜欢高兴的猫。 [That dog likes the happy cat.]*
Student:	**C.** 那只狗喜欢高兴的猫。
	Ok! Then I believe you forgot to conjugate the verb ‘to like’. Also, remember that an adjective must come before the noun it’s modifying. Please try again!

* The English translation is what the system thinks is correct, but it is not shown.

Summary

- **Prescriptive Grammars**

  (We're not looking for full coverage)

- **Graded Lexicons & Graded Parsers**

  (From surveying educational materials)

- **Implementation Mal-Rules ('error production rules')**

  (From Learner Corpora)

- **Random Exercises (e.g. build a sentence, …)**

  (Controlled for both grammar and lexicon levels)

- **Pairwise Semantic-based Translation / Disambiguation**

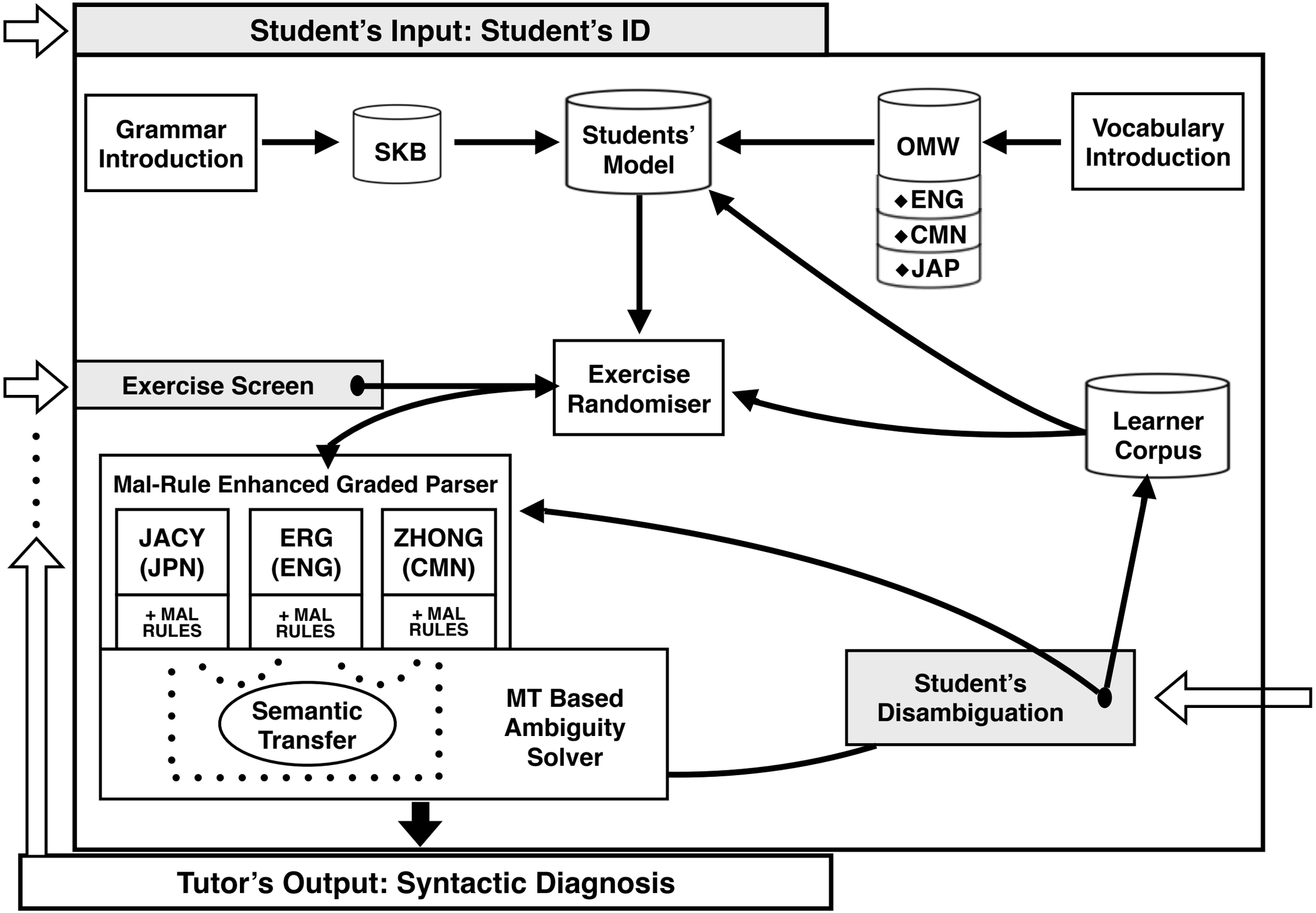
  (Eng ↔ Cmn, Eng ↔ Jpn, Cmn↔Jpn, to pinpoint errors)

- **Automatic collection Learner corpora**

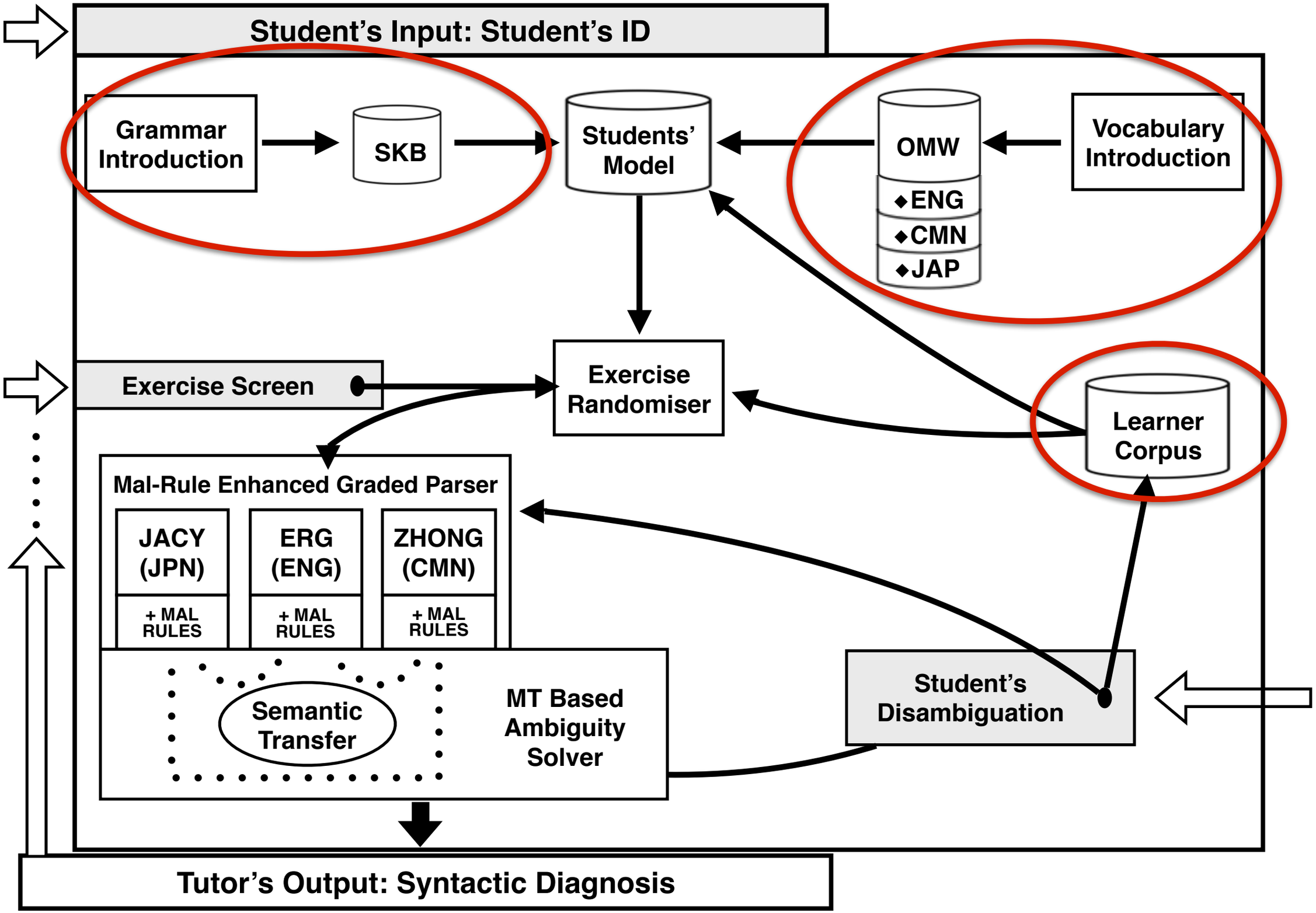
  (Ungrammatical input + tree-banked, sense-tagged corrected sentence)

- **Controlled experiment using NTU students of Mandarin L2**

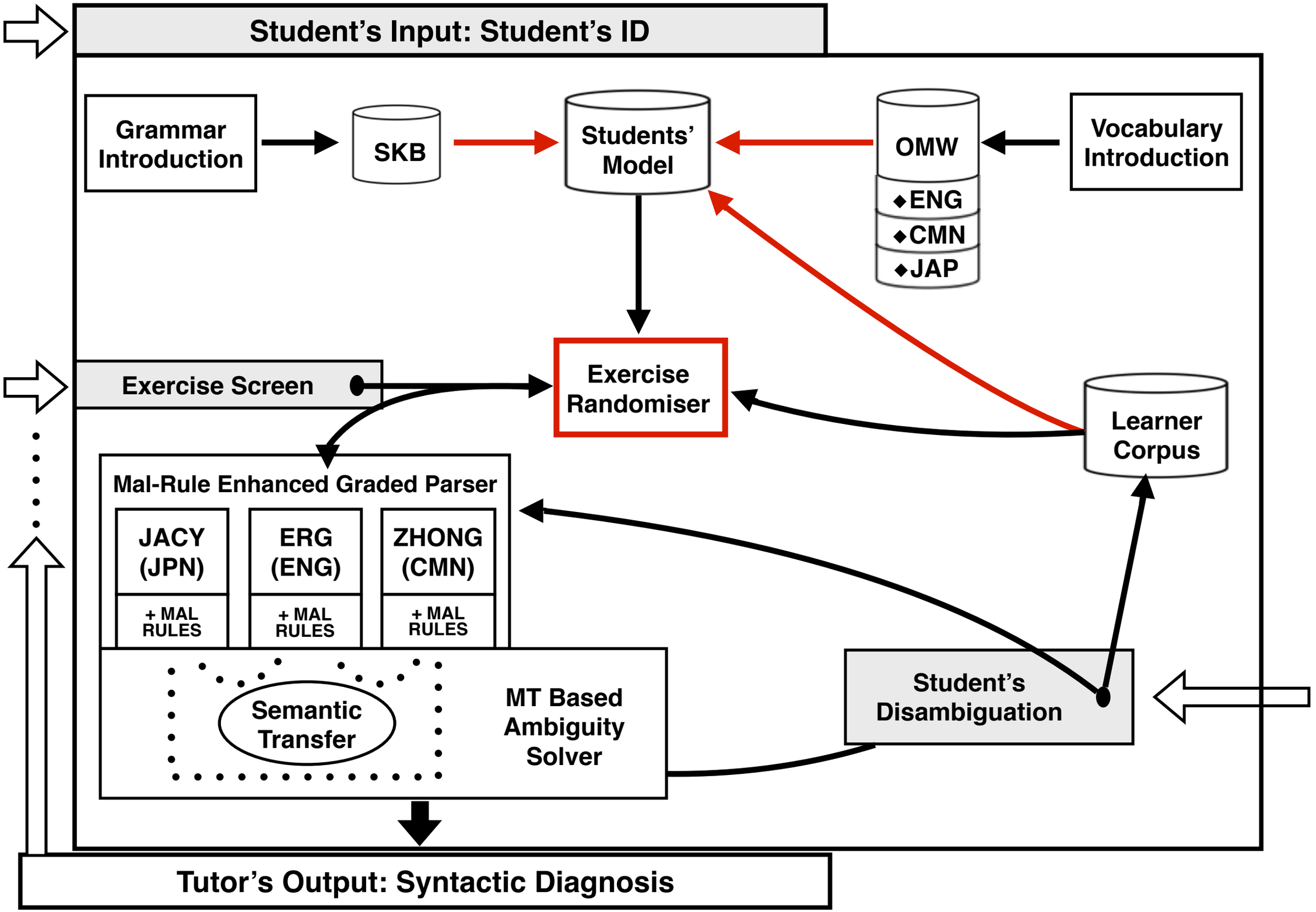
System's Flowchart



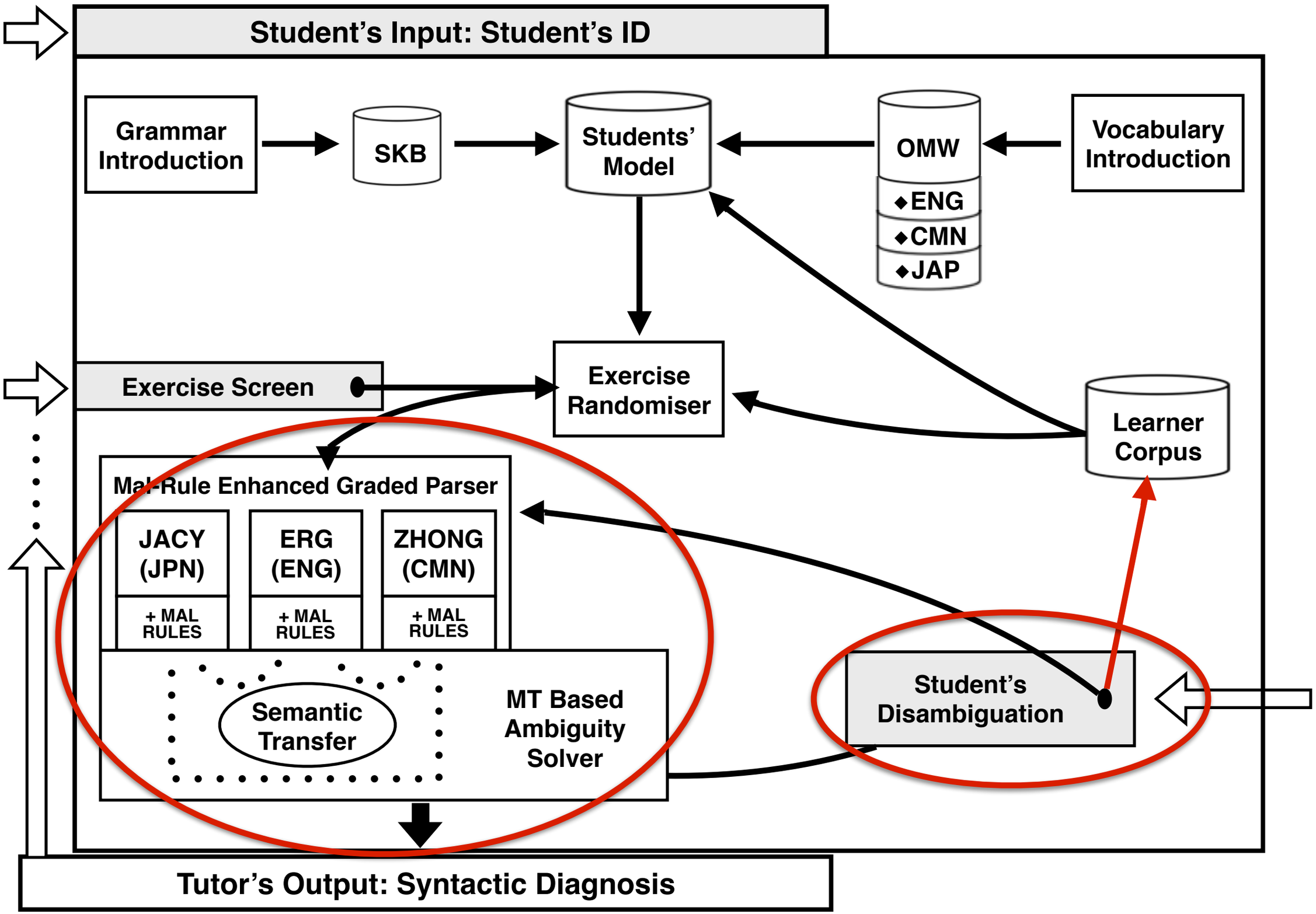
System's Flowchart



System's Flowchart



System's Flowchart



If you think this is cool, if you want to know more, if have any ideas to improve this, or if you foresee any problem with our approach… Let's talk about it!

Thank You!